

Using CAN Devices / 使用 CAN 设备

CAN has many advantages over other methods of connection between the robot controller and peripheral devices.

与机器人控制器和外围设备之间的其他连接方式相比，CAN 具有许多优势。

- CAN connections are daisy-chained from device to device, which often results in much shorter wire runs than having to wire each device to the RIO itself.
- CAN 连接以菊花链方式从一个设备连接到另一个设备，因此与必须将每个设备分别连接到 RIO 相比，通常可以显著缩短线缆长度。
- Much more data can be sent over a CAN connection than over a PWM connection - thus, CAN motor controllers are capable of a much more expansive feature-set than are PWM motor controllers.
- CAN 连接能够传输的数据远多于 PWM 连接，因此 CAN 电机控制器可以提供比 PWM 电机控制器丰富得多的功能集。
- CAN is bi-directional, so CAN motor controllers can send data back to the RIO, again facilitating a more expansive feature-set than can be offered by PWM Controllers.
- CAN 是双向的，因此 CAN 电机控制器能够将数据发回 RIO，这同样使其能够提供比 PWM 控制器更丰富的功能集。

For instructions on wiring CAN devices, see the relevant section of the robot wiring guide.

有关 CAN 设备的接线说明，请参阅 WPILib 文档中机器人接线指南的相关章节。

CAN devices generally have their own WPILib classes. The following sections will describe the use of several of these classes.

CAN 设备通常具有各自对应的 WPILib 类。以下章节将介绍其中若干类的用法。

Pneumatics Control Module / Pneumatics Control Module



The Pneumatics Control Module (PCM) is a CAN-based device that provides complete control over the compressor and up to 8 solenoids per module. The PCM is integrated into WPILib through a series of classes that make it simple to use.

Pneumatics Control Module (PCM) 是一种基于 CAN 的设备，可以完整控制压缩机以及每个模块上最多 8 个电磁阀。PCM 通过一系列类集成到 WPILib 中，因而使用起来十分简单。

The closed loop control of the Compressor and Pressure switch is handled by the `Compressor` class ([Java](#)), and the Solenoids are handled by the `Solenoid` ([Java](#)) and `DoubleSolenoid` ([Java](#)) classes.

压缩机与压力开关的闭环控制由 `Compressor` 类 ([Java](#)) 处理，电磁阀则由 `Solenoid` ([Java](#)) 和 `DoubleSolenoid` ([Java](#)) 类处理。

An additional PCM module can be used where the module's corresponding solenoids are differentiated by the module number in the constructors of the `Solenoid` and `Compressor` classes.

可以使用额外的 PCM 模块；在 `Solenoid` 和 `Compressor` 类的构造函数中，通过模块编号区分各模块对应的电磁阀。

For more information on controlling the compressor, see [Operating a Compressor for Pneumatics](#) in the WPILib documentation.

有关控制压缩机的更多信息，请参阅 WPILib 文档中的“Operating a Compressor for Pneumatics”。

For more information on controlling solenoids, see Operating Pneumatic Cylinders in the WPILib documentation.

有关控制电磁阀的更多信息，请参阅 WPILib 文档中的“Operating Pneumatic Cylinders”。

Pneumatic Hub / Pneumatic Hub



The Pneumatic Hub (PH) is a CAN-based device that provides complete control over the compressor and up to 16 solenoids per module. The PH is integrated into WPILib through a series of classes that make it simple to use.

Pneumatic Hub (PH) 是一种基于 CAN 的设备，可以完整控制压缩机以及每个模块上最多 16 个电磁阀。PH 通过一系列类集成到 WPILib 中，因而使用起来十分简单。

The closed loop control of the Compressor and Pressure switch is handled by the `Compressor` class ([Java](#)), and the Solenoids are handled by the `Solenoid` ([Java](#)) and `DoubleSolenoid` ([Java](#)) classes.

压缩机与压力开关的闭环控制由 `Compressor` 类 ([Java](#)) 处理，电磁阀则由 `Solenoid` ([Java](#)) 和 `DoubleSolenoid` ([Java](#)) 类处理。

An additional PH module can be used where the module's corresponding solenoids are differentiated by the module number in the constructors of the Solenoid and Compressor classes.

可以使用额外的 PH 模块；在 `Solenoid` 和 `Compressor` 类的构造函数中，通过模块编号区分各模块对应的电磁阀。

For more information on controlling the compressor, see Operating a Compressor for Pneumatics in the WPILib documentation.

有关控制压缩机的更多信息，请参阅 WPILib 文档中的“Operating a Compressor for Pneumatics”。

For more information on controlling solenoids, see Operating Pneumatic Cylinders in the WPILib documentation.

有关控制电磁阀的更多信息，请参阅 WPILib 文档中的“Operating Pneumatic Cylinders”。

Power Distribution Module / 配电模块

The CTRE Power Distribution Panel (PDP) and Rev Power Distribution Hub (PDH) can use their CAN connectivity to communicate a wealth of status information regarding the robot's power use to the roboRIO, for use in user code. This has the capability to report current temperature, the bus voltage, the total robot current draw, the total robot energy use, and the individual current draw of each device power channel. This data can be used for a number of advanced control techniques, such as motor torque limiting and brownout avoidance.

CTRE Power Distribution Panel (PDP) 和 REV Power Distribution Hub (PDH) 可以利用 CAN 连接，将大量与机器人用电情况有关的状态信息传送到 roboRIO，供用户代码使用。它们可以报告当前温度、总线电压、机器人总电流消耗、机器人总能量消耗，以及每个设备供电 Channel 的单独电流消耗。这些数据可以用于多种高级控制技术，例如限制电机扭矩和避免欠压。

Creating a Power Distribution Object / 创建 Power Distribution 对象

To use the either Power Distribution module, create an instance of the `PowerDistribution` class ([Java](#)). With no arguments, the Power Distribution object will be detected, and must use CAN ID of 0 for CTRE or 1 for REV. If the CAN ID is non-default, additional constructors are available to specify the CAN ID and type.

要使用任一种配电模块，请创建 `PowerDistribution` 类 ([Java](#)) 的实例。不传入参数时，将自动检测 Power Distribution 对象，并且 CTRE 设备必须使用 CAN ID 0，REV 设备必须使用 CAN ID 1。如果 CAN ID 不是默认值，可以使用其他构造函数指定 CAN ID 和类型。

```
PowerDistribution examplePD = new PowerDistribution();
PowerDistribution examplePD = new PowerDistribution(0, ModuleType.kCTRE);
PowerDistribution examplePD = new PowerDistribution(1, ModuleType.kRev);
```

NOTE: it is not necessary to create a `PowerDistribution` object unless you need to read values from it. The board will work and supply power on all the channels even if the object is never created.

说明：除非需要从配电模块读取数值，否则无需创建 `PowerDistribution` 对象。即使从未创建该对象，配电板仍会正常工作，并向所有 Channel 供电。

Reading the Bus Voltage / 读取总线电压

```
// Get the voltage going into the PDP, in volts.
// The PDP returns the voltage in increments of 0.05 volts.
double voltage = m_pdp.getVoltage();
SmartDashboard.putNumber("voltage", voltage);
```

Monitoring the bus voltage can be useful for (among other things) detecting when the robot is near a brownout, so that action can be taken to avoid brownout in a controlled manner. See the roboRIO Brownouts document in the WPILib documentation for more information.

监测总线电压的用途之一是检测机器人何时接近欠压，从而能够采取措施，以受控方式避免欠压。有关更多信息，请参阅 WPILib 文档中的“roboRIO Brownouts”。

Reading the Temperature / 读取温度

```
// Retrieves the temperature of the PDP, in degrees Celsius.  
double temperatureCelsius = m_pdp.getTemperature();  
SmartDashboard.putNumber("Temperature", temperatureCelsius);
```

Monitoring the temperature can be useful for detecting if the robot has been drawing too much power and needs to be shut down for a while, or if there is a short or other wiring problem.

监测温度有助于检测机器人是否消耗了过多功率、是否需要停机一段时间，或者是否存在短路或其他接线问题。

Reading the Total Current, Power, and Energy / 读取总电流、功率和能量

```
// Get the total current of all channels.  
double totalCurrent = m_pdp.getTotalCurrent();  
SmartDashboard.putNumber("Total Current", totalCurrent);  
  
// Get the total power of all channels.  
// Power is the bus voltage multiplied by the current with the units Watts.  
double totalPower = m_pdp.getTotalPower();  
SmartDashboard.putNumber("Total Power", totalPower);  
  
// Get the total energy of all channels.  
// Energy is the power summed over time with units Joules.  
double totalEnergy = m_pdp.getTotalEnergy();  
SmartDashboard.putNumber("Total Energy", totalEnergy);
```

Monitoring the total current, power, and energy can be useful for controlling how much power is being drawn from the battery, both for preventing brownouts and ensuring that mechanisms have sufficient power available to perform the actions required. Power is the bus voltage multiplied by the current with the units Watts. Energy is the power summed over time with units Joules.

监测总电流、功率和能量有助于控制从电池获取的功率，既可以防止欠压，也可以确保机构具有足够的可用功率来执行所需动作。功率等于总线电压乘以电流，单位为 Watts。能量是功率随时间的累加值，单位为 Joules。

Reading Individual Channel Currents / 读取各个 Channel 的电流

The PDP/PDH also allows users to monitor the current drawn by the individual device power channels. You can read the current on any of the 16 PDP channels (0-15) or 24 PDH channels (0-23).

PDP/PDH 还允许用户监测各个设备供电 Channel 消耗的电流。可以读取 PDP 的任意一个 16 个 Channel (0-15) 或 PDH 的任意一个 24 个 Channel (0-23) 上的电流。

```
// Get the current going through channel 7, in Amperes.  
// The PDP returns the current in increments of 0.125A.  
// At low currents the current readings tend to be less accurate.  
double current7 = m_pdp.getCurrent(7);  
SmartDashboard.putNumber("Current Channel 7", current7);
```

Monitoring individual device current draws can be useful for detecting shorts or stalled motors.

监测各个设备的电流消耗有助于检测短路或电机堵转。

Using the Switchable Channel (PDH) / 使用可开关 Channel (PDH)

The REV PDH has one channel that can be switched on or off to control custom circuits.

REV PDH 具有一个可以开启或关闭的 Channel，用于控制自定义电路。

```
examplePD.setSwitchableChannel(true);  
examplePD.setSwitchableChannel(false);
```

Third-Party CAN Devices / 第三方 CAN 设备

A number of FRC® vendors offer their own CAN peripherals. As CAN devices offer expansive feature-sets, vendor CAN devices require similarly expansive code libraries to operate. As a result, these libraries are not maintained as an official part of WPILib, but are instead maintained by the vendors themselves. For a guide to installing third-party libraries, see [3rd Party Libraries](#)

许多 FRC® 厂商都提供自己的 CAN 外围设备。由于 CAN 设备具有丰富的功能集，厂商的 CAN 设备也需要功能同样丰富的代码库才能运行。因此，这些库并非作为 WPILib 的正式组成部分进行维护，而是由各厂商自行维护。有关安装第三方库的指南，请参阅[“3rd Party Libraries”](#)。

A list of common third-party CAN devices from various vendors, along with links to corresponding external documentation, is provided below:

下面列出了来自不同厂商的常见第三方 CAN 设备，并提供了相应外部文档的链接：

CTR Electronics / CTR Electronics

CTR Electronics (CTRE) offers several CAN peripherals with external libraries. General resources for all CTRE devices include:

CTR Electronics（CTRE）提供多种配有外部库的 CAN 外围设备。适用于所有 CTRE 设备的通用资料包括：

[Phoenix Device Software Documentation](#)

[Phoenix Device 软件文档](#)

CTRE Motor Controllers / CTRE 电机控制器

- **Talon FX (with Falcon 500 Motor, Kraken x60 motor, and Kraken x44)**

Talon FX（搭配 Falcon 500 Motor、Kraken x60 motor 和 Kraken x44）

- API Documentation (v6: [Java](#))
API 文档（v6: [Java](#)）
- Hardware User's Manual ([Falcon 500](#), [Kraken x60](#), [Kraken x44](#))
硬件用户手册（[Falcon 500](#)、[Kraken x60](#)、[Kraken x44](#)）
- [Software Documentation](#)
[软件文档](#)

- **Talon SRX**

Talon SRX

- API Documentation ([Java](#))
API 文档（[Java](#)）
- [Hardware User's Manual](#)
[硬件用户手册](#)
- [Software Documentation](#)
[软件文档](#)

- **Victor SPX**

Victor SPX

- API Documentation ([Java](#))
API 文档 ([Java](#))
- [Hardware User's Manual](#)
[硬件用户手册](#)
- [Software Documentation](#)
[软件文档](#)

CTRE Sensors / CTRE 传感器

- **CANcoder**

CANcoder

- API Documentation (v6: [Java](#))
API 文档 (v6: [Java](#))
- [Hardware User's Manual](#)
[硬件用户手册](#)
- [Software Documentation](#)
[软件文档](#)

- **Pigeon 2.0**

Pigeon 2.0

- API Documentation (v6: [Java](#))
API 文档 (v6: [Java](#))
- [Hardware User's Manual](#)
[硬件用户手册](#)
- [Software Documentation6](#)
[软件文档6](#)

- **Pigeon IMU**

Pigeon IMU

- API Documentation ([Java](#))
API 文档 ([Java](#))
- [Hardware User's Manual](#)
[硬件用户手册](#)
- [Software Documentation](#)
[软件文档](#)

- **CANifier**

CANifier

- API Documentation ([Java](#))

API 文档 ([Java](#))

- [Hardware User's Manual](#)

[硬件用户手册](#)

- [Software Documentation](#)

[软件文档](#)

CTRE Other CAN Devices / CTRE 其他 CAN 设备

- **CANdle LED Controller**

CANdle LED 控制器

- API Documentation ([Java](#))

API 文档 ([Java](#))

- [Hardware User's Manual](#)

[硬件用户手册](#)

- [Software Documentation](#)

[软件文档](#)

REV Robotics / REV Robotics

REV Robotics currently offers the SPARK MAX and SPARK Flex motor controllers which can be used for brushed and REV brushless (NEO, NEO 550, and NEO Vortex) motors.

REV Robotics 目前提供 SPARK MAX 和 SPARK Flex 电机控制器，可用于有刷电机和 REV 无刷电机（NEO、NEO 550 和 NEO Vortex）。

REV Motor Controllers / REV 电机控制器

- **SPARK MAX**

SPARK MAX

- API Documentation ([Java](#))

API 文档 ([Java](#))

- [Technical Manual](#)

[技术手册](#)

- **SPARK Flex**

SPARK Flex

- API Documentation ([Java](#))

API 文档 ([Java](#))

- [Technical Manual](#)

[技术手册](#)

Playing With Fusion / Playing With Fusion

Playing With Fusion (PWF) offers the Venom integrated motor/controller as well as a Time-of-Flight distance sensor:

Playing With Fusion (PWF) 提供 Venom 一体式电机/控制器以及 Time-of-Flight 测距传感器：

PWF Motor Controllers / PWF 电机控制器

- **Venom**

- Venom**

- [API Documentation \(Java\)](#)

- API 文档 ([Java](#))

- [Technical Manual](#)

- [技术手册](#)

PWF Sensors / PWF 传感器

- **Time of Flight Sensor**

- Time of Flight 传感器**

- [API Documentation \(Java\)](#)

- API 文档 ([Java](#))

- [Technical Manual](#)

- [技术手册](#)

Redux Robotics / Redux Robotics

WARNING / 警告： Redux Robotics is shutting down. While their devices remain legal for FRC use and software support will be provided for the 2026 season, no new hardware will be available.

Redux Robotics 正在停止运营。尽管其设备仍可合法用于 FRC，并且会在 2026 赛季提供软件支持，但将不再提供新硬件。

Redux Robotics offers the HELIUM Canandmag CAN + PWM magnetic encoder and the BORON Canandgyro CAN-enabled gyro.

Redux Robotics 提供 HELIUM Canandmag CAN + PWM 磁性编码器，以及支持 CAN 的 BORON Canandgyro 陀螺仪。

Redux Sensors / Redux 传感器

- **HELIUM Canandmag**

- HELIUM Canandmag**

- [API Documentation \(Java\)](#)

- API 文档 ([Java](#))

- [Technical Manual](#)

- [技术手册](#)

- **BORON Canandgyro**

BORON Canandgyro

- API Documentation ([Java](#))
API 文档 ([Java](#))
- [Technical Manual](#)
[技术手册](#)

Grapple Robotics / Grapple Robotics

Grapple Robotics currently offers the LaserCAN CAN-enabled range finding sensor

Grapple Robotics 目前提供支持 CAN 的 LaserCAN 测距传感器。

Grapple Sensors / Grapple 传感器

- **LaserCAN**

LaserCAN

- API Documentation ([Java](#))
API 文档 ([Java](#))
- [Technical Manual](#)
[技术手册](#)

FRC CAN Device Specifications / FRC CAN 设备规范

This document seeks to describe the basic functions of the current FRC® CAN system and the requirements for any new CAN devices seeking to work with the system.

本文档旨在说明当前 FRC® CAN 系统的基本功能，以及希望与该系统配合使用的任何新 CAN 设备必须满足的要求。

Addressing / 寻址

FRC CAN nodes assign arbitration IDs based on a pre-defined scheme that breaks the ID into 5 components:

FRC CAN 节点按照预定义方案分配仲裁 ID，该方案将 ID 分为 5 个组成部分：

Device Type / 设备类型

This is a 5-bit value describing the type of device being addressed. A table of currently assigned device types can be found below. If you wish to have a new device type assigned from the `Reserved` pool, please submit a request to FIRST.

这是一个 5-bit 值，用于描述当前寻址的设备类型。当前已分配的设备类型见下表。如果希望从 `Reserved` 池中分配新的设备类型，请向 FIRST 提交申请。

Device Types	Value
Broadcast Messages	0
Robot Controller	1
Motor Controller	2
Relay Controller	3
Gyro Sensor	4
Accelerometer	5
Distance Sensor	6
Encoder	7
Power Distribution Module	8
Pneumatics Controller	9
Miscellaneous	10
IO Breakout	11
Servo Controller	12
Color Sensor	13
Reserved	14-30

Device Types	Value
Firmware Update	31

设备类型	值
广播消息	0
Robot Controller	1
电机控制器	2
继电器控制器	3
陀螺仪传感器	4
加速度计	5
距离传感器	6
编码器	7
配电模块	8
气动控制器	9
其他设备	10
IO Breakout	11
舵机控制器	12
颜色传感器	13
保留	14-30
固件更新	31

Manufacturer / 制造商

This is an 8-bit value indicating the manufacturer of the CAN device. Currently assigned values can be found in the table below. If you wish to have a manufacturer ID assigned from the **Reserved** pool, please submit a request to FIRST.

这是一个 8-bit 值，用于表示 CAN 设备的制造商。当前已分配的值见下表。如果希望从 **Reserved** 池中分配制造商 ID，请向 FIRST 提交申请。

Manufacturer	Value
Broadcast	0
NI	1
Luminary Micro	2

Manufacturer	Value
DEKA	3
CTR Electronics	4
REV Robotics	5
Grapple	6
MindSensors	7
Team Use	8
Kauai Labs	9
Copperforge	10
Playing With Fusion	11
Studica	12
The Thrifty Bot	13
Redux Robotics	14
AndyMark	15
Vivid Hosting	16
Vertos Robotics	17
SWYFT Robotics	18
Lumyn Labs	19
Brushland Labs	20
Reserved	21-255

制造商	值
广播	0
NI	1
Luminary Micro	2
DEKA	3
CTR Electronics	4
REV Robotics	5
Grapple	6
MindSensors	7

制造商	值
队伍使用	8
Kauai Labs	9
Copperforge	10
Playing With Fusion	11
Studica	12
The Thrifty Bot	13
Redux Robotics	14
AndyMark	15
Vivid Hosting	16
Vertos Robotics	17
SWYFT Robotics	18
Lumyn Labs	19
Brushland Labs	20
保留	21-255

API/Message Identifier / API/消息标识符

The API or Message Identifier is a 10-bit value that identifies a particular command or message type. These identifiers are unique for each Manufacturer + Device Type combination (so an API identifier that may be a "Voltage Set" for a Luminary Micro Motor Controller may be a "Status Get" for a CTR Electronics Motor Controller or `Current Get` for a CTR Power Distribution Module).

API 或消息标识符是一个 10-bit 值，用于标识特定的命令或消息类型。对于每一种 Manufacturer + Device Type 组合，这些标识符都是唯一的（因此，某个 API 标识符对于 Luminary Micro Motor Controller 可能表示“Voltage Set”，对于 CTR Electronics Motor Controller 可能表示“Status Get”，而对于 CTR Power Distribution Module 则可能表示 `Current Get`）。

The Message identifier is further broken down into 2 sub-fields: the 6-bit API Class and the 4-bit API Index.
消息标识符还会进一步分为 2 个子字段：6-bit API Class 和 4-bit API Index。

API Class / API Class

The API Class is a 6-bit identifier for an API grouping. Similar messages are grouped into a single API Class. An example of the API Classes for a CAN Motor Controller is shown in the table below.

API Class 是一个 6-bit 标识符，用于表示 API 分组。相似的消息会归入同一个 API Class。下表给出了 CAN Motor Controller 的 API Class 示例。

API Class	Value
Voltage Control Mode	0
Speed Control Mode	1
Voltage Compensation Mode	2
Position Control Mode	3
Current Control Mode	4
Status	5
Periodic Status	6
Configuration	7
Ack	8

API Class	值
电压控制模式	0
速度控制模式	1
电压补偿模式	2
位置控制模式	3
电流控制模式	4
状态	5
周期状态	6
配置	7
Ack	8

API Index / API Index

The API Index is a 4-bit identifier for a particular message within an API Class. An example of the API Index values for a CAN Motor Controller Speed Control API Class is shown in the table below.

API Index 是一个 4-bit 标识符，用于标识 API Class 中的特定消息。下表给出了 CAN Motor Controller 的 Speed Control API Class 中各 API Index 值的示例。

API Index	Value
Enable Control	0
Disable Control	1
Set Setpoint	2

API Index	Value
P Constant	3
I Constant	4
D Constant	5
Set Reference	6
Trusted Enable	7
Trusted Set No Ack	8
Trusted Set Setpoint No Ack	10
Set Setpoint No Ack	11

API Index	值
启用控制	0
禁用控制	1
设置 setpoint	2
P 常数	3
I 常数	4
D 常数	5
设置 reference	6
可信启用	7
可信设置且无需 Ack	8
可信设置 setpoint 且无需 Ack	10
设置 setpoint 且无需 Ack	11

Device Number / 设备编号

Device Number is a 6-bit quantity indicating the number of the device of a particular type. Devices should default to device ID 0 to match other components of the FRC Control System. Device 0x3F may be reserved for device specific broadcast messages.

Device Number 是一个 6-bit 量，用于表示某一特定类型设备的编号。设备的默认 device ID 应为 0，以便与 FRC Control System 的其他组件保持一致。设备 0x3F 可以保留用于设备专用的广播消息。

Example
Speed Control Mode Disable from Luminary Micro Jaguar Speed Controller (dev # 4)

Field Value	Device Type					Manufacturer Code								API							Device Number								
	2					2								API Class							Index				4				
Bits	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	
Bit Position	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Protected Frames / 受保护帧

FRC CAN Nodes which implement actuator control capability (motor controllers, relays, pneumatics controllers, etc.) must implement a way to verify that the robot is enabled and that commands originate with the main robot controller (i.e. the roboRIO).

实现执行器控制能力的 FRC CAN 节点（电机控制器、继电器、气动控制器等）必须实现一种机制，用于验证机器人处于启用状态，并且命令来自主机器人控制器（即 roboRIO）。

Broadcast Messages / 广播消息

Broadcast messages are messages sent to all nodes by setting the device type and manufacturer fields to 0. The API Class for broadcast messages is 0. The currently defined broadcast messages are shown in the table below:

广播消息是通过将 device type 和 manufacturer 字段设为 0 而发送给所有节点的消息。广播消息的 API Class 为 0。当前定义的广播消息见下表：

Description	Value
Disable	0
System Halt	1
System Reset	2
Device Assign	3
Device Query	4
Heartbeat	5
Sync	6
Update	7
Firmware Version	8
Enumerate	9
System Resume	10

描述	值
禁用	0
系统停止	1
系统复位	2
设备分配	3
设备查询	4
心跳	5
同步	6
更新	7
固件版本	8
枚举	9
系统恢复	10

Devices should disable immediately when receiving the Disable message (arbID 0). Implementation of other broadcast messages is optional.

设备收到 Disable 消息（arbID 0）后应立即禁用。其他广播消息的实现是可选的。

Requirements for FRC CAN Nodes / FRC CAN 节点要求

For CAN Nodes to be accepted for use in the FRC System, they must:

CAN 节点若要获准用于 FRC System，必须满足以下要求：

- Communicate using Arbitration IDs which match the prescribed FRC format:
使用符合规定 FRC 格式的 Arbitration ID 进行通信：
 - A valid, issued CAN Device Type (per Table 1 - CAN Device Types)
有效且已分配的 CAN Device Type（参见表 1——CAN Device Types）
 - A valid, issued Manufacturer ID (per Table 2 - CAN Manufacturer Codes)
有效且已分配的 Manufacturer ID（参见表 2——CAN Manufacturer Codes）
 - API Class(es) and Index(s) assigned and documented by the device manufacturer
由设备制造商分配并记录的 API Class 和 Index
 - A user selectable device number if multiple units of the device type are intended to co-exist on the same network.
如果同一设备类型的多个设备需要共存于同一网络，则必须提供用户可选择的设备编号。
- Support the minimum Broadcast message requirements as detailed in the Broadcast Messages section.
支持“Broadcast Messages”章节中详述的最低广播消息要求。
- If controlling actuators, utilize a scheme to assure that the robot is issuing commands, is enabled, and is still present.

如果控制执行器，则采用一种机制，以确保机器人正在发出命令、处于启用状态且仍然在线。

- Provide software library support for Java or arrange with *FIRST*® or FIRST's Control System Partners to provide such interfaces.

提供 Java 软件库支持，或者与 *FIRST*® 或 FIRST 的 Control System Partner 协调，由其提供此类接口。

Universal Heartbeat / 通用心跳

The roboRIO provides a universal CAN heartbeat that any device on the bus can listen and react to. This heartbeat is sent every 20 ms. The heartbeat has a full CAN ID of 0x01011840 (which is the NI Manufacturer ID, RobotController type, Device ID 0 and API ID 0x061). It is an 8 byte CAN packet with the following bitfield layout.

roboRIO 提供通用 CAN 心跳，总线上的任何设备都可以监听并对其作出响应。该心跳每 20 ms 发送一次。此心跳的完整 CAN ID 为 0x01011840（其中包含 NI Manufacturer ID、RobotController 类型、Device ID 0 和 API ID 0x061）。它是一个 8 byte 的 CAN 数据包，具有如下位字段布局。

Description	Byte	Width (bits)
Match time (seconds)	8	8
Match number	6-7	10
Replay number	6	6
Red alliance	5	1
Enabled	5	1
Autonomous mode	5	1
Test mode	5	1
System watchdog	5	1
Tournament type	5	3
Time of day (year)	4	6
Time of day (month)	3-4	4
Time of day (day)	3	5
Time of day (seconds)	2-3	6
Time of day (minutes)	1-2	6
Time of day (hours)	1	5

描述	Byte	宽度 (bit)
比赛时间（秒）	8	8
比赛编号	6-7	10

描述	Byte	宽度 (bit)
重赛编号	6	6
红方联盟	5	1
已启用	5	1
Autonomous 模式	5	1
Test 模式	5	1
System watchdog	5	1
赛事类型	5	3
当前时间 (年)	4	6
当前时间 (月)	3-4	4
当前时间 (日)	3	5
当前时间 (秒)	2-3	6
当前时间 (分)	1-2	6
当前时间 (时)	1	5

If the `system watchdog` flag is set, motor controllers are enabled. If 100 ms has passed since this packet was received, the robot program can be considered hung, and devices should act as if the robot has been disabled.

如果设置了 `system watchdog` 标志，则电机控制器处于启用状态。如果自收到该数据包以来已经过去 100 ms，则可以认为机器人程序已经挂起，设备应按照机器人已被禁用的状态运行。

Note that all fields except `Enabled`, `Autonomous mode`, `Test mode`, and `system watchdog` will contain invalid values until an arbitrary time after the Driver Station connects.

请注意，在 Driver Station 连接后的某个不确定时间点之前，除 `Enabled`、`Autonomous mode`、`Test mode` 和 `system watchdog` 外的所有字段都将包含无效值。